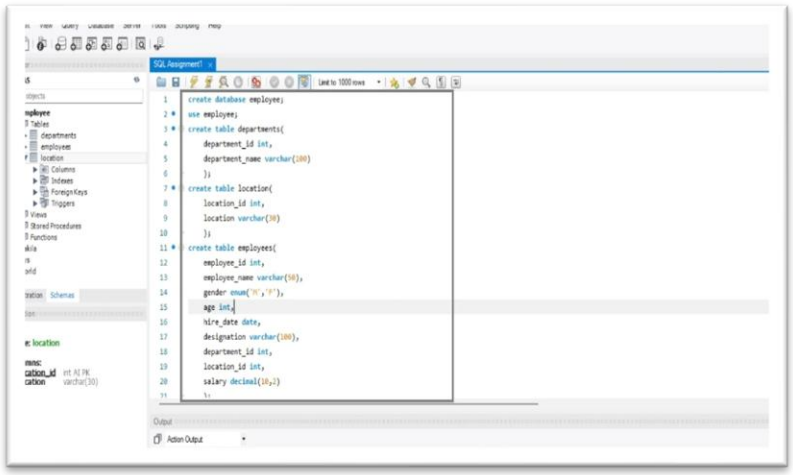
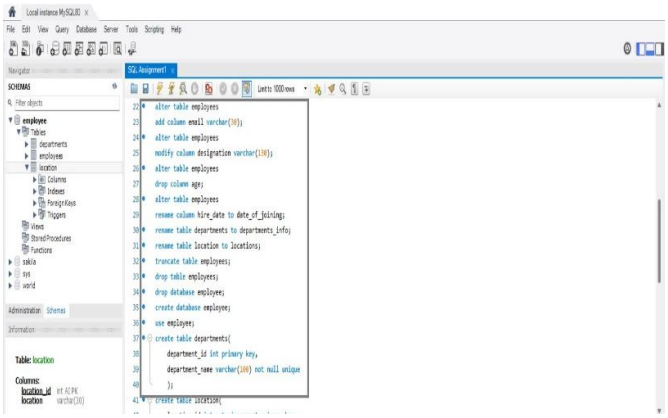
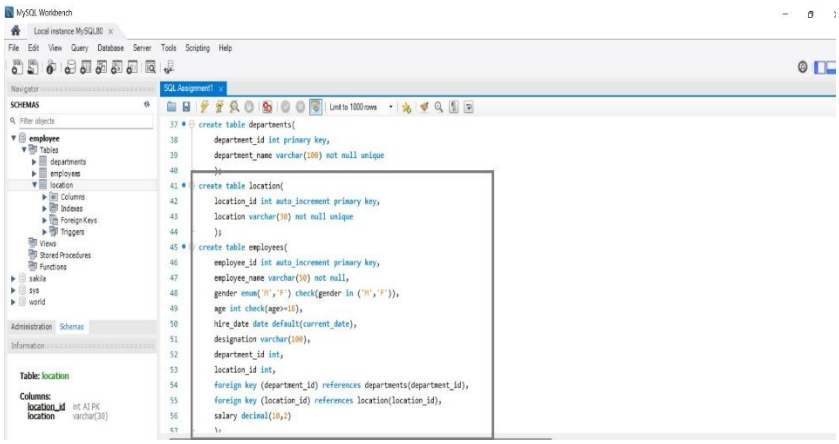
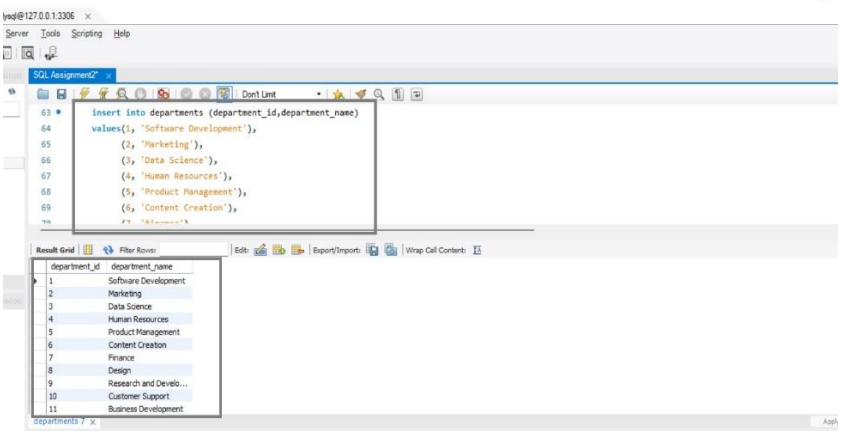

SQL Assignment1
create database employee;
use employee;
create table departments(
department_id int,
department_name varchar(100)
);
create table location(
location_id int,
location varchar(30)
);
create table employees(
employee_id int,
employee_name varchar(50),
gender enum('M','F'),
age int,
hire_date date,
designation varchar(100),
department_id int,
location_id int,
salary decimal(10,2)
);
Output
Action Output

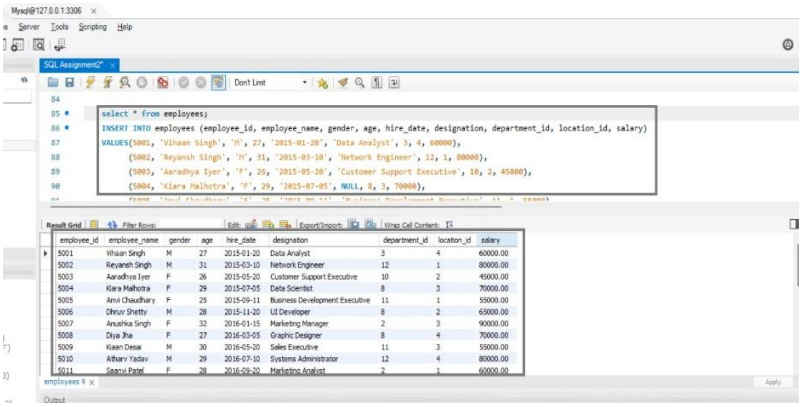

Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
SQL Assignment1
SCHEMAS
employee
Tables
departments
employees
location
Columns
Indexes
Foreign Keys
Triggers
Views
Stored Procedures
Functions
sakila
sys
world
Administration Schemas
Information
Table: location
Columns:
location_id int AI PK
location varchar(30)
alter table employees
add column email varchar(30);
alter table employees
modify column designation varchar(130);
alter table employees
drop column age;
alter table employees
rename column hire_date to date_of_joining;
rename table departments to departments_info;
rename table location to locations;
truncate table employees;
drop table employees;
drop database employee;
create database employee;
use employee;
create table departments(
department_id int primary key,
department_name varchar(100) not null unique
);


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
SQL Assignment1
SCHEMAS
employee
Tables
departments
employees
location
Columns
Indexes
Foreign Keys
Triggers
Views
Stored Procedures
Functions
sakila
sys
world
Administration Schemas
Information
Table: location
Columns:
location_id int AI PK
location varchar(30)
create table departments(
department_id int primary key,
department_name varchar(100) not null unique
);
create table location(
location_id int auto_increment primary key,
location varchar(30) not null unique
);
create table employees(
employee_id int auto_increment primary key,
employee_name varchar(50) not null,
gender enum('M','F') check(gender in ('M','F')),
age int check(age>18),
hire_date date default(current_date),
designation varchar(100),
department_id int,
location_id int,
foreign key (department_id) references departments(department_id),
foreign key (location_id) references location(location_id),
salary decimal(10,2)
);



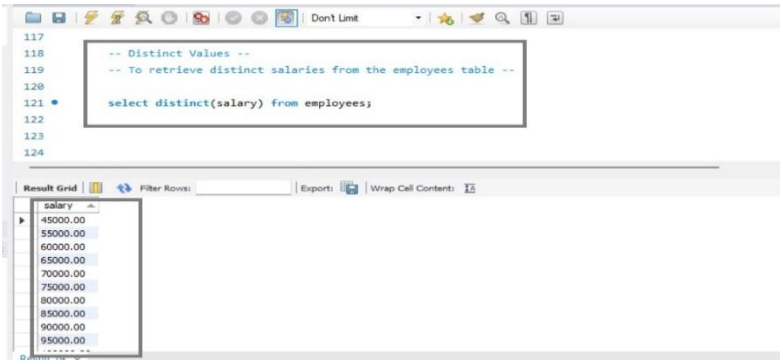
```
insert into departments (department_id,department_name)
values(1, 'Software Development'),
      (2, 'Marketing'),
      (3, 'Data Science'),
      (4, 'Human Resources'),
      (5, 'Product Management'),
      (6, 'Content Creation'),
```

department_id	department_name
1	Software Development
2	Marketing
3	Data Science
4	Human Resources
5	Product Management
6	Content Creation
7	Finance
8	Design
9	Research and Develo...
10	Customer Support
11	Business Development



```
select * from employees;
INSERT INTO employees (employee_id, employee_name, gender, age, hire_date, designation, department_id, location_id, salary)
VALUES(5001, 'Vihaan Singh', 'M', 27, '2015-01-20', 'Data Analyst', 3, 4, 60000),
      (5002, 'Reyansh Singh', 'M', 31, '2015-03-10', 'Network Engineer', 12, 1, 80000),
      (5003, 'Aaradhya Iyer', 'F', 26, '2015-05-20', 'Customer Support Executive', 10, 2, 45000),
      (5004, 'Kiara Malhotra', 'F', 29, '2015-07-05', NULL, 8, 3, 70000),
```

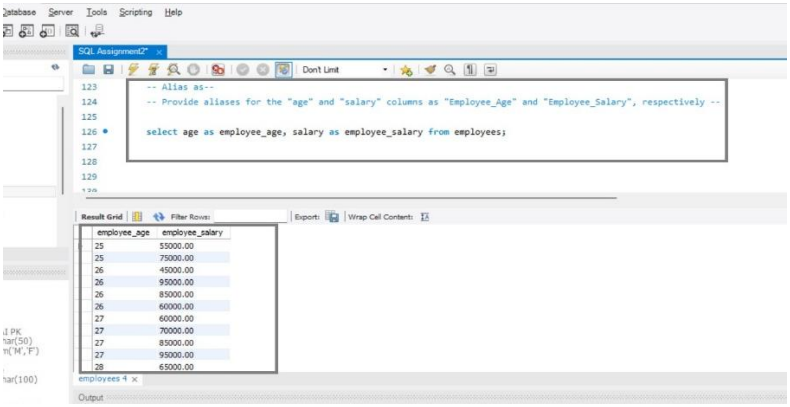
employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000.00
5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	12	1	80000.00
5003	Aaradhya Iyer	F	26	2015-05-20	Customer Support Executive	10	2	45000.00
5004	Kiara Malhotra	F	29	2015-07-05	Data Scientist	8	3	70000.00
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	11	1	55000.00
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	8	2	65000.00
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	2	3	90000.00
5008	Diya Jha	F	27	2016-03-05	Graphic Designer	8	4	70000.00
5009	Kiaan Desai	M	30	2016-05-20	Sales Executive	11	3	55000.00
5010	Atharv Yadav	M	29	2016-07-10	Systems Administrator	12	4	80000.00
5011	Saanvi Patel	F	28	2016-09-20	Marketing Analyst	2	1	60000.00



```
-- Distinct Values --
-- To retrieve distinct salaries from the employees table --

select distinct(salary) from employees;
```

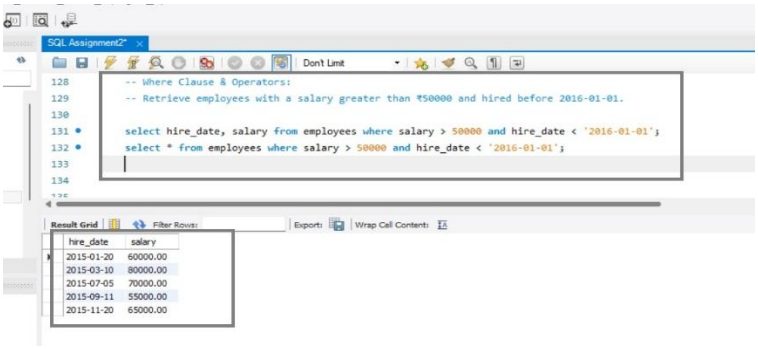
salary
45000.00
55000.00
60000.00
65000.00
70000.00
75000.00
80000.00
85000.00
90000.00
95000.00



```
-- Alias as--
-- Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively --

select age as employee_age, salary as employee_salary from employees;
```

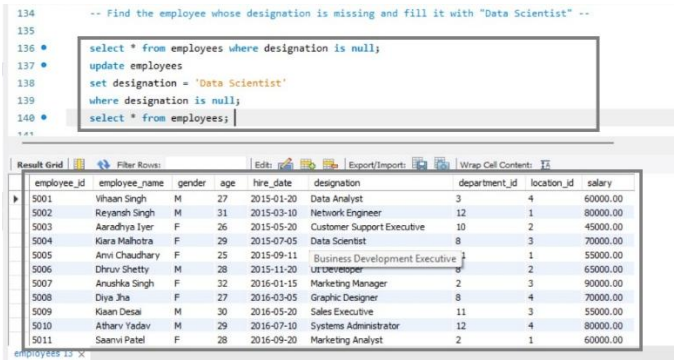
employee_age	employee_salary
25	55000.00
25	75000.00
26	45000.00
26	95000.00
26	85000.00
26	60000.00
27	60000.00
27	70000.00
27	85000.00
27	95000.00
28	65000.00



```
-- Where Clause & Operators:
-- Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

select hire_date, salary from employees where salary > 50000 and hire_date < '2016-01-01';
select * from employees where salary > 50000 and hire_date < '2016-01-01';
```

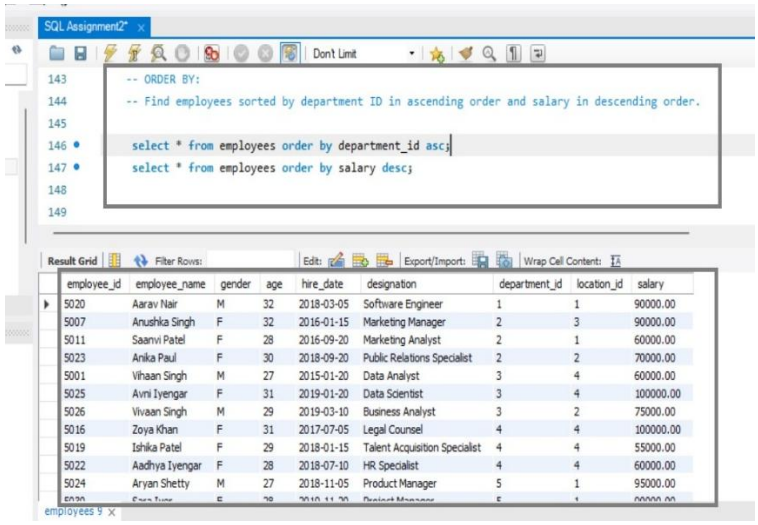
hire_date	salary
2015-01-20	60000.00
2015-03-10	80000.00
2015-07-05	70000.00
2015-09-11	55000.00
2015-11-20	65000.00



```
-- Find the employee whose designation is missing and fill it with "Data Scientist" --

select * from employees where designation is null;
update employees
set designation = 'Data Scientist'
where designation is null;
select * from employees;
```

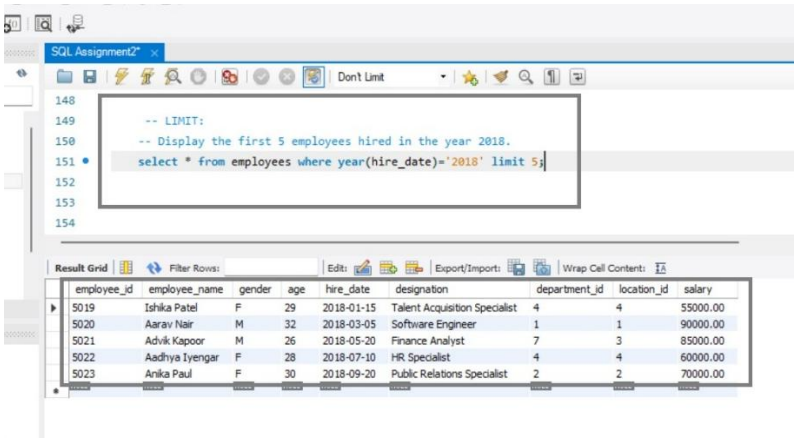
employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000.00
5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	12	1	80000.00
5003	Aaradhya Iyer	F	26	2015-05-20	Customer Support Executive	10	2	45000.00
5004	Kiara Malhotra	F	29	2015-07-05	Data Scientist	8	3	70000.00
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	3	1	55000.00
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	8	2	65000.00
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	2	3	90000.00
5008	Diya Jha	F	27	2016-03-05	Graphic Designer	8	4	70000.00
5009	Kiaan Desai	M	30	2016-05-20	Sales Executive	11	3	55000.00
5010	Atharv Yadav	M	29	2016-07-10	Systems Administrator	12	4	80000.00
5011	Saanvi Patel	F	28	2016-09-20	Marketing Analyst	2	1	60000.00

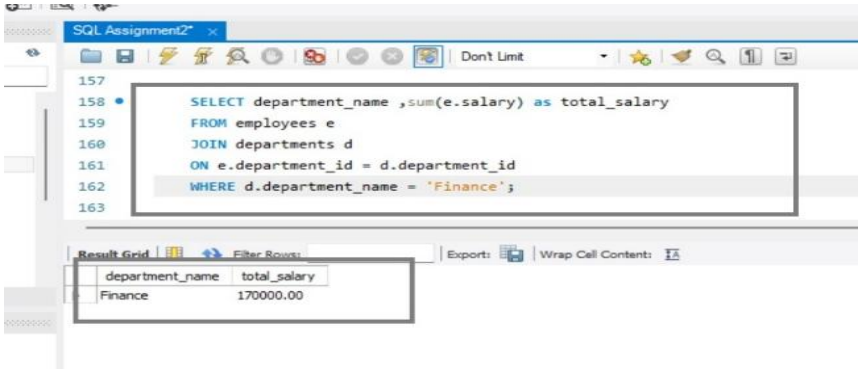


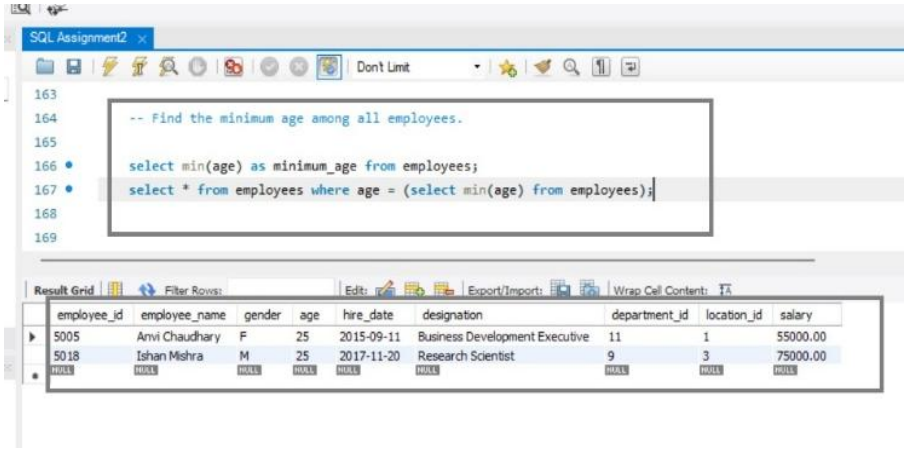
```
-- ORDER BY:
-- Find employees sorted by department ID in ascending order and salary in descending order.

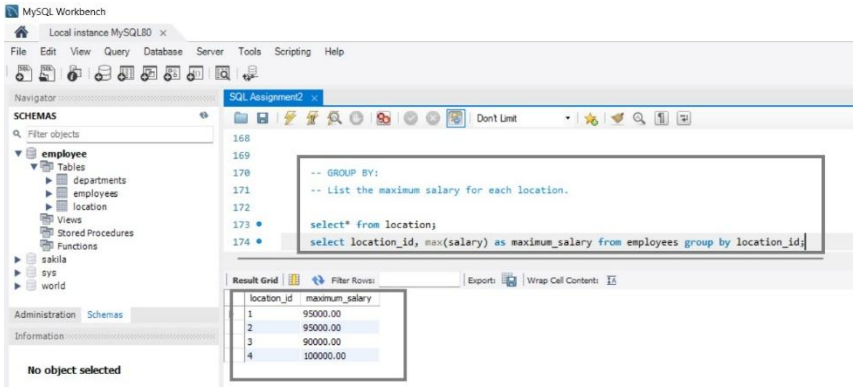
select * from employees order by department_id asc;
select * from employees order by salary desc;
```

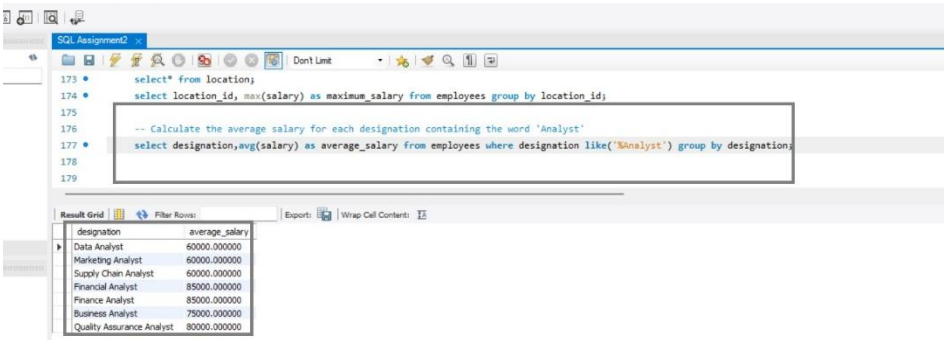
employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	1	1	90000.00
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	2	3	90000.00
5011	Saanvi Patel	F	28	2016-09-20	Marketing Analyst	2	1	60000.00
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	2	2	70000.00
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000.00
5025	Avni Iyengar	F	31	2019-01-20	Data Scientist	3	4	100000.00
5026	Vivaan Singh	M	29	2019-03-10	Business Analyst	3	2	75000.00
5016	Zoya Khan	F	31	2017-07-05	Legal Counsel	4	4	100000.00
5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	4	4	55000.00
5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	4	4	60000.00
5024	Aryan Shetty	M	27	2018-11-05	Product Manager	5	1	95000.00

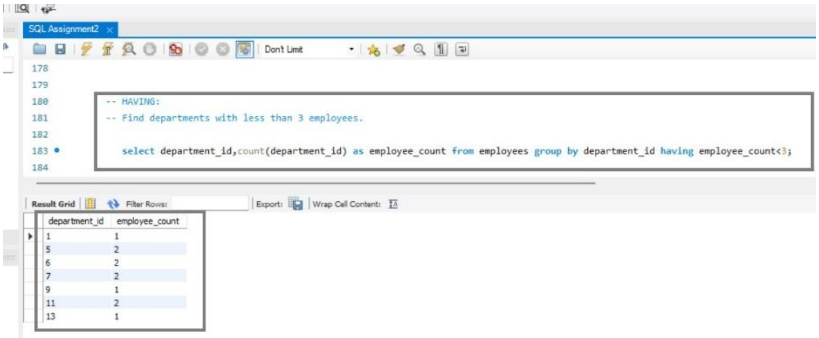

SQL Assignment2*
Don't Limit
148
149 -- LIMIT:
150 -- Display the first 5 employees hired in the year 2018.
151 • select * from employees where year(hire_date)='2018' limit 5;
152
153
154
Result Grid
Filter Rows:
Edit:
Export/Import:
Wrap Cell Content:
employee_id employee_name gender age hire_date designation department_id location_id salary
5019 Ishika Patel F 29 2018-01-15 Talent Acquisition Specialist 4 4 55000.00
5020 Aarav Nair M 32 2018-03-05 Software Engineer 1 1 90000.00
5021 Advik Kapoor M 26 2018-05-20 Finance Analyst 7 3 85000.00
5022 Aadhya Iyengar F 28 2018-07-10 HR Specialist 4 4 60000.00
5023 Anika Paul F 30 2018-09-20 Public Relations Specialist 2 2 70000.00

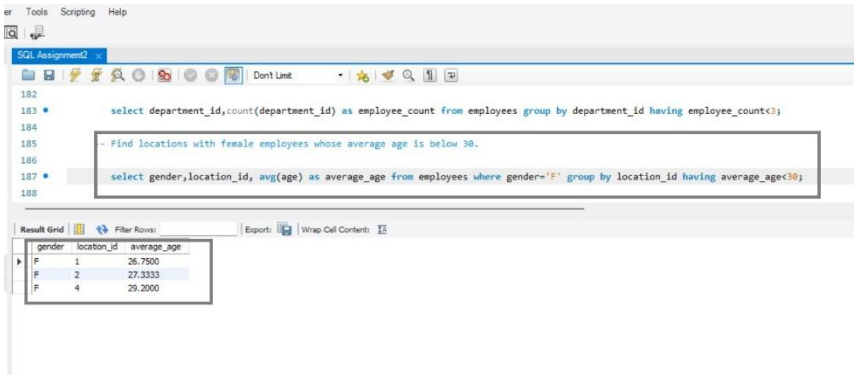

SQL Assignment2*
Don't Limit
157
158 • SELECT department_name ,sum(e.salary) as total_salary
159 FROM employees e
160 JOIN departments d
161 ON e.department_id = d.department_id
162 WHERE d.department_name = 'Finance';
163
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
department_name total_salary
Finance 170000.00

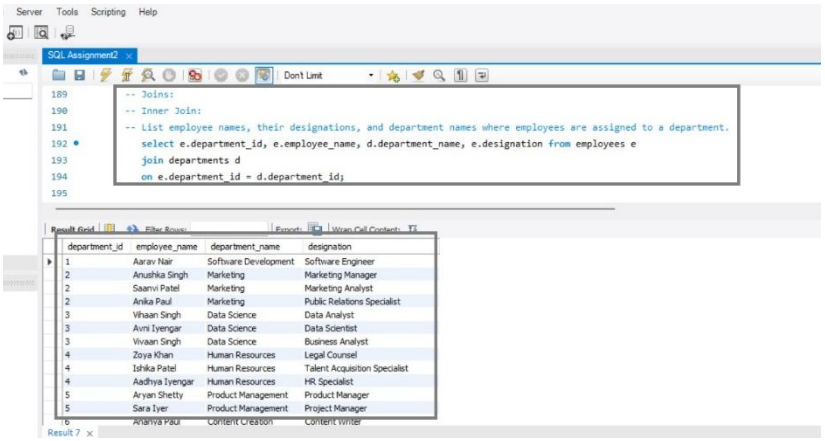

SQL Assignment2
Don't Limit
163
164 -- Find the minimum age among all employees.
165
166 • select min(age) as minimum_age from employees;
167 • select * from employees where age = (select min(age) from employees);
168
169
Result Grid
Filter Rows:
Edit:
Export/Import:
Wrap Cell Content:
employee_id employee_name gender age hire_date designation department_id location_id salary
5005 Anvi Chaudhary F 25 2015-09-11 Business Development Executive 11 1 55000.00
5018 Ishan Mishra M 25 2017-11-20 Research Scientist 9 3 75000.00

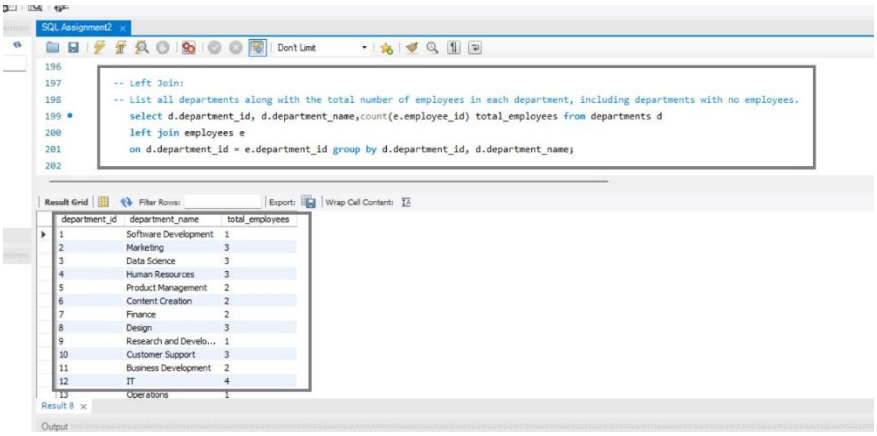

MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
employee
Tables
departments
employees
location
Views
Stored Procedures
Functions
sakila
sys
world
Administration Schemas
Information
No object selected
SQL Assignment2
Don't Limit
168
169
170 -- GROUP BY:
171 -- List the maximum salary for each location.
172
173 select* from location;
174 select location_id, max(salary) as maximum_salary from employees group by location_id;
Result Grid Filter Rows: Export: Wrap Cell Content:
location_id maximum_salary
1 95000.00
2 95000.00
3 90000.00
4 100000.00

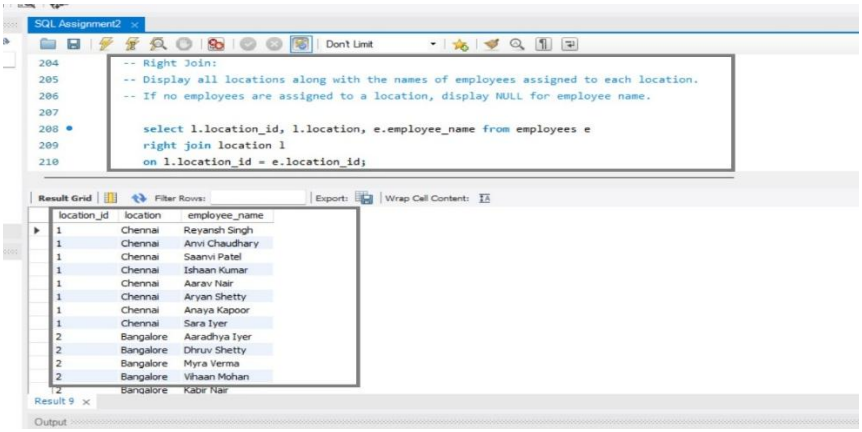

SQL Assignment2
Don't Limit
173 select* from location;
174 select location_id, max(salary) as maximum_salary from employees group by location_id;
175
176 -- Calculate the average salary for each designation containing the word 'Analyst'
177 select designation,avg(salary) as average_salary from employees where designation like('%Analyst') group by designation;
178
179
Result Grid Filter Rows: Export: Wrap Cell Content:
designation average_salary
Data Analyst 60000.000000
Marketing Analyst 60000.000000
Supply Chain Analyst 60000.000000
Financial Analyst 85000.000000
Finance Analyst 85000.000000
Business Analyst 75000.000000
Quality Assurance Analyst 80000.000000


SQL Assignment2
Don't Limit
178
179
180 -- HAVING:
181 -- Find departments with less than 3 employees.
182
183 select department_id,count(department_id) as employee_count from employees group by department_id having employee_count<3;
184
Result Grid Filter Rows: Export: Wrap Cell Content:
department_id employee_count
1 1
5 2
6 2
7 2
9 1
11 2
13 1


er Tools Scripting Help
SQL Assignment2
Don't Limit
182
183 select department_id,count(department_id) as employee_count from employees group by department_id having employee_count<3;
184
185 - Find locations with female employees whose average age is below 30.
186
187 select gender,location_id, avg(age) as average_age from employees where gender='F' group by location_id having average_age<30;
188
Result Grid Filter Rows: Export: Wrap Cell Content:
gender location_id average_age
F 1 26.7500
F 2 27.3333
F 4 29.2000


Server Tools Scripting Help
SQL Assignment2
Don't Limit
-- Joins:
-- Inner Join:
-- List employee names, their designations, and department names where employees are assigned to a department.
select e.department_id, e.employee_name, d.department_name, e.designation from employees e
join departments d
on e.department_id = d.department_id;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
department_id employee_name department_name designation
1 Aarav Nair Software Development Software Engineer
2 Anushka Singh Marketing Marketing Manager
2 Saanvi Patel Marketing Marketing Analyst
2 Anika Paul Marketing Public Relations Specialist
3 Vihaan Singh Data Science Data Analyst
3 Avni Iyengar Data Science Data Scientist
3 Vivaan Singh Data Science Business Analyst
4 Zoya Khan Human Resources Legal Counsel
4 Ishika Patel Human Resources Talent Acquisition Specialist
4 Aadhya Iyengar Human Resources HR Specialist
5 Aryan Shetty Product Management Product Manager
5 Sara Iyer Product Management Project Manager
Result 7


SQL Assignment2
Don't Limit
-- Left Join:
-- List all departments along with the total number of employees in each department, including departments with no employees.
select d.department_id, d.department_name,count(e.employee_id) total_employees from departments d
left join employees e
on d.department_id = e.department_id group by d.department_id, d.department_name;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
department_id department_name total_employees
1 Software Development 1
2 Marketing 3
3 Data Science 3
4 Human Resources 3
5 Product Management 2
6 Content Creation 2
7 Finance 2
8 Design 3
9 Research and Develo... 1
10 Customer Support 3
11 Business Development 2
12 IT 4
13 Operations 1
Result 8
Output


SQL Assignment2
Don't Limit
-- Right Join:
-- Display all locations along with the names of employees assigned to each location.
-- If no employees are assigned to a location, display NULL for employee name.
select l.location_id, l.location, e.employee_name from employees e
right join location l
on l.location_id = e.location_id;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
location_id location employee_name
1 Chennai Reyansh Singh
1 Chennai Anvi Chaudhary
1 Chennai Saanvi Patel
1 Chennai Ishaan Kumar
1 Chennai Aarav Nair
1 Chennai Aryan Shetty
1 Chennai Anaya Kapoor
1 Chennai Sara Iyer
2 Bangalore Aaradhya Iyer
2 Bangalore Dhruv Shetty
2 Bangalore Myra Verma
2 Bangalore Vihaan Mohan
2 Bangalore Kabir Nair
Result 9
Output